GD&T Question Bank

30 Questions with Graded Difficulty

Each question: **10 marks**

Suggested working time: **25–30 minutes per question**

Level I:	Foundation and interpretation (Questions 1–10)
Level II:	Applied analysis and assembly (Questions 11–20)
Level III:	Advanced synthesis and inspection (Questions 21–30)

Prepared for classroom tests, tutorials, and examinations

Instructions and conventions

Use ASME Y14.5-2018 (R2024) conventions unless a question states a simplifying assumption. All dimensions are in millimetres. Basic dimensions are theoretically exact. Unless stated otherwise, a quoted measured feature size is its actual mating size and all local sizes satisfy their limits. For coordinate-based position calculations, assume the stated offsets describe a straight axis parallel to its nominal direction unless tilt is specified. Show the principal steps for every numerical decision and state *pass* or *fail*. Equality with a tolerance limit passes; compare unrounded results. A quoted required position-zone diameter for an axis is evaluated about true position in the established datum reference frame, using only permitted common pattern or datum mobility; do not best-fit holes separately. The boxed feature-control frames use geometric characteristic symbols. A circled M after a tolerance denotes MMC; after a datum letter it denotes MMB.

- A diametrical position error based on coordinate offsets is

$$T_{\text{pos}} = 2\sqrt{(\Delta x)^2 + (\Delta y)^2}.$$

- Internal feature at MMC: bonus = actual mating size − MMC size.
- External feature at MMC: bonus = MMC size − actual mating size.
- Unless specifically stated otherwise, ignore measurement uncertainty, surface texture, elastic deformation, and thermal expansion.

Difficulty and coverage map

Q.	Principal topic	Dominant skill	Level
1	Position at MMC	Interpretation and acceptance	I
2	Flatness and line straightness	Minimum-zone calculations	I
3	Axis straightness at MMC	Bonus tolerance and virtual condition	I
4	Axis perpendicularity	Orientation-zone analysis	I
5	Datum precedence	Constraint and set-up reasoning	I
6	Rule 1 and circularity	Size-form interaction	I
7	Unequally disposed profile zone	Boundary and process reasoning	I
8	Circular and total runout	Indicator-data interpretation	I
9	Functional gaging of a hole pattern	Virtual-condition decisions	I
10	Basic versus plus/minus dimensions	Drawing consistency	I
11	Datum shift at MMB	Boundary mobility and acceptance	II
12	Projected tolerance zone	Fastener-axis projection	II
13	Composite position	Pattern location and refinement	II
14	Composite versus stacked controls	Specification comparison	II
15	Fixed-fastener assembly	Worst-case tolerance allocation	II
16	Floating-fastener assembly	Relative pattern displacement	II
17	Simultaneous requirement	Common datum set-up	II
18	Datum targets on a casting	Constrained alignment and profile	II
19	Runout diagnosis	Circumferential and axial error	II
20	Coaxial bores	Position versus runout	II
21	Sequential datum shift	Mixed slot and cylindrical datums	III
22	Composite position with varying sizes	Per-feature bonus assessment	III
23	Functional gage design	Simulators and virtual boundaries	III
24	Assembly over a temperature range	Functional tolerance budgeting	III
25	Profile and machining stock	Feasible-process interval	III
26	Position of a slot	Non-cylindrical feature of size	III
27	Multiple datum reference frames	Functional relationship and inspection	III
28	Conformance and process capability	Normalized tolerance utilization	III
29	Critique of an over-controlled drawing	Error diagnosis and redesign	III
30	Bearing-housing capstone	Complete GD&T scheme synthesis	III

Question 1: Position at MMC

Level I

A plate contains a through-hole of size ⌀12.00–⌀12.25. Its location is defined by basic dimensions and controlled by

⌖	⌀0.20 Ⓜ	A	B	C

.

The measured hole size is ⌀12.16, and the axis is displaced from true position by $\Delta x = +0.12$ and $\Delta y = -0.10$.

(a) Describe the shape, size, and orientation of the specified tolerance zone. **[2 marks]**

(b) Determine the hole MMC, LMC, bonus tolerance, and total available position tolerance. **[2 marks]**

(c) Calculate the diametrical true-position error from the coordinate deviations. **[3 marks]**

(d) Decide whether the hole conforms and state the maximum permissible radial axis displacement at the measured size. **[2 marks]**

(e) Explain why the dimensions locating the hole should be basic rather than plus/minus toleranced. **[1 mark]**

Question 2: Flatness and straightness

Level I

A machined face is controlled by flatness 0.070. The middle longitudinal line element is additionally controlled by straightness 0.035. A CMM reports signed normal deviations at four equally spaced x-stations on three equally spaced longitudinal lines. The deviations are reported in the respective minimum-zone orientations for the sampled surface and the middle line y_2. Treat these samples as the complete data for this exercise.

	Signed normal deviations (mm)			
Line	x_1	x_2	x_3	x_4
y_1	$+0.040$	$+0.031$	$+0.026$	-0.021
y_2	-0.006	$+0.022$	$+0.040$	-0.006
y_3	-0.021	-0.004	$+0.015$	$+0.040$

(a) Calculate the minimum separation of two parallel planes required to contain all measured points and assess flatness. **[3 marks]**

(b) Calculate the straightness error of the middle line element y_2 and assess it. **[2 marks]**

(c) Explain how the surface can pass one control and fail the other. **[2 marks]**

(d) State whether either control requires a datum and justify your answer. **[1 mark]**

(e) Recommend one additional inspection path that would reduce the risk of missing a local form defect. **[2 marks]**

Question 3: Axis straightness at MMC

Level I

A shaft has size ∅19.90–∅20.00 and the derived median line is controlled by

–	∅0.08 Ⓜ

.

For one part, the actual mating size is ∅19.94, and the smallest cylindrical zone that contains the derived median line has diameter 0.125.

(a) Identify the shaft MMC and calculate the bonus straightness tolerance. **[2 marks]**

(b) Determine the total available straightness tolerance and assess the part. **[2 marks]**

(c) Calculate the virtual-condition diameter of the shaft. **[2 marks]**

(d) Reassess the same measured axis error if the actual mating size were ∅19.99. **[2 marks]**

(e) Distinguish axis straightness from straightness of a surface line element. **[2 marks]**

Question 4: Perpendicularity of a hole axis

Level I

A ∅8.00–∅8.25 hole passes through a 30-mm-thick bracket. Its axis is controlled by

⊥	∅0.12 Ⓜ	A

.

The measured hole size is ∅8.18. The CMM determines that a cylindrical zone of diameter 0.24, oriented perpendicular to datum A, is required to contain the complete derived axis.

(a) Calculate the bonus and total perpendicularity tolerance. **[2 marks]**

(b) Decide whether the feature conforms. **[2 marks]**

(c) Estimate the corresponding angular error if a straight axis has an end-to-end lateral displacement of 0.24 over the thickness. **[2 marks]**

(d) Reassess conformance if the hole size were only ∅8.05. **[2 marks]**

(e) Explain why this orientation control does not, by itself, locate the hole axis in the plane of datum A. **[2 marks]**

Question 5: Datum precedence and constraints

Level I

A rectangular component uses the bottom face as datum A, a long side face as datum B, and an adjacent end face as datum C. Let A nominally coincide with the xy plane, B with the xz plane, and C with the yz plane.

(a) List the six rigid-body degrees of freedom and identify which are constrained by primary datum A. **[3 marks]**

(b) Identify the additional degrees of freedom constrained by secondary datum B and tertiary datum C. **[2 marks]**

(c) Describe a practical 3–2–1 inspection fixture for establishing this datum reference frame. **[2 marks]**

(d) Explain how interchanging B and C can change inspection results when the two side faces are not perfectly perpendicular. **[2 marks]**

(e) State one functional reason for selecting the long side as the secondary datum. **[1 mark]**

Question 6: Rule 1 and circularity

Level I

A shaft is specified as ∅24.90–∅25.00 with no general independency note. CMM inspection gives local diameters between ∅24.93 and ∅24.99, but the minimum circumscribed cylinder enclosing the complete surface has diameter ∅25.03. At the worst cross-section, the minimum-zone circularity error is 0.038.

(a) Determine whether the local size measurements satisfy the size limits. **[2 marks]**

(b) Apply the perfect-form-at-MMC envelope requirement and decide whether the shaft conforms. **[3 marks]**

(c) If circularity 0.040 were separately specified, assess that control. **[2 marks]**

(d) Explain why passing circularity would not cancel the envelope violation. **[2 marks]**

(e) Assuming a circular local section concentric with the envelope, state the diametrical and radial clearance from a local ∅24.94 section to the ∅25.00 MMC envelope. **[1 mark]**

Question 7: Unequally disposed profile

Level I

A cast surface is controlled by surface profile 0.80 relative to $A|B|C$. The zone is explicitly disposed so that 0.20 lies outside the true profile and 0.60 lies inside it. Positive deviation denotes excess material outside the true profile. Inspection gives a maximum deviation of $+0.17$ and a minimum deviation of -0.54. For the coating questions, assume the same profile requirement applies to the coated surface and the coating thickness is uniform in the local surface-normal direction.

(a) State the upper and lower profile boundaries. **[2 marks]**

(b) Assess the as-cast surface. **[2 marks]**

(c) A coating adds a uniform outward thickness of 0.10. Determine the new extreme deviations and reassess the part. **[2 marks]**

(d) Calculate the maximum uniform coating thickness that can be added without violating the profile requirement. **[2 marks]**

(e) Determine whether the original surface would pass an equally disposed profile tolerance of 0.80. **[2 marks]**

Level I

Question 8: Circular and total runout

Radial indicator readings are taken with one unchanged indicator zero while a cylindrical land is rotated about datum axis A. The values capture the extrema at the two sampled axial sections; they do not establish what happens between those sections.

	0°	45°	90°	135°	180°	225°	270°	315°
Z_1	+.012	+.027	+.041	+.033	+.006	−.009	−.014	−.001
Z_2	+.020	+.031	+.045	+.038	+.016	+.001	−.003	+.008

(a) Calculate circular runout at each section and assess a circular-runout tolerance of 0.050. **[4 marks]**

(b) Calculate the observed overall indicator range across the two measured sections and compare it with a total-runout tolerance of 0.060. **[2 marks]**

(c) Explain why the readings must not be independently recentered at each section when datum axis A is specified. **[2 marks]**

(d) Name one geometric error that can contribute to circular runout, and state what additional inspection is needed before declaring the complete land conforming for total runout. **[2 marks]**

Question 9: Functional gaging of holes

Level I

Four holes have size ∅10.00–∅10.25 and position

⌖	∅0.25 Ⓜ	A	B	C

.

CMM results, all evaluated in the same datum reference frame, are:

Hole	Actual size (mm)	Required position-zone diameter (mm)
1	10.04	0.22
2	10.12	0.34
3	10.18	0.47
4	10.25	0.50

(a) Calculate the hole virtual condition and the corresponding fixed-gage pin diameter. **[2 marks]**

(b) For each hole, calculate the available position tolerance and state pass or fail. **[6 marks]**

(c) State the acceptance decision for the complete four-hole pattern. **[1 mark]**

(d) Explain the functional meaning of bonus tolerance in this application. **[1 mark]**

Question 10: Basic and coordinate dimensions

Level I

A drawing locates a ⌀10.00–⌀10.20 hole at basic coordinates $X = 50$ and $Y = 30$, controls its position by ⌀0.20 at MMC relative to $A|B|C$, and also shows 50 ± 0.05 and 30 ± 0.05 for the same coordinates. The measured hole size is ⌀10.10, and its centre is at $(50.08, 29.94)$.

(a) Identify the specification conflict created by the two coordinate schemes. **[2 marks]**

(b) Calculate the true-position error and assess the hole using the positional tolerance. **[3 marks]**

(c) Assess the same location using the plus/minus coordinate dimensions. **[2 marks]**

(d) Rewrite the dimensioning strategy so that the location is controlled unambiguously. **[2 marks]**

(e) Explain why rectangular coordinate limits do not define the same acceptance boundary as a cylindrical position zone. **[1 mark]**

Question 11: Datum shift at MMB

Level II

A controlled hole H has size ⌀10.00–⌀10.25 and position

⌖	⌀0.20 Ⓜ	A	B Ⓜ	C

.

Datum feature B is an internal cylindrical hole of size ⌀20.00–⌀20.20. For this problem, its maximum-material boundary is ⌀20.00. The actual mating size of B is ⌀20.14. Hole H measures ⌀10.10 and has coordinate offsets $\Delta x = 0.18$, $\Delta y = 0.06$. Assume the available datum shift can act in the most favourable radial direction.

(a) Calculate the total position tolerance available from the size of hole H alone. **[2 marks]**

(b) Calculate the maximum radial datum shift available from datum feature B. **[2 marks]**

(c) Assess the controlled hole without using datum shift. **[2 marks]**

(d) Apply the maximum useful shift, calculate the residual radial and diametrical position errors, and reassess the feature. **[3 marks]**

(e) State why datum shift is not an additional manufacturing error allowance on datum feature B. **[1 mark]**

Question 12: Projected tolerance zone

Level II

A threaded hole in an 8-mm-thick bracket must accept a long stud. The mounting face is datum A, and basic dimensions locate the hole from B and C. Its straight axis is controlled by the schematic projected-zone frame

⌖	∅0.30 Ⓟ 25	A	B	C

.

The cylindrical zone projects 25 above the mounting face. At that face, the axis has an offset vector $(0.04, 0.03)$. Due to tilt, the additional displacement over the projected height is $(0.16, -0.04)$. Assume a straight axis, so its radial offset varies linearly over the projection.

(a) Calculate the resultant radial axis displacement at the top of the projected zone. **[3 marks]**

(b) Determine whether the axis lies inside the projected tolerance cylinder. **[2 marks]**

(c) If the axis were centred at the mounting face, calculate the maximum permissible angular tilt over the 25-mm projection. **[2 marks]**

(d) If a radial offset of 0.05 already exists at the mounting face and tilt occurs in the same direction, calculate the remaining permissible tilt angle. **[2 marks]**

(e) State the functional problem that a projected tolerance zone is intended to prevent. **[1 mark]**

Question 13: Composite position

Level II

A four-hole pattern has hole size ⌀10.00–⌀10.20 and the composite control

⌖	⌀0.40 Ⓜ	A	B	C
	⌀0.15 Ⓜ	A		

.

All four holes initially measure ⌀10.10. From one admissible simultaneous fit of the common upper and lower frameworks to all four holes, the CMM reports that the upper pattern-locating zones each require at most ⌀0.46, while the lower feature-relating zones each require at most ⌀0.22. Do not fit each hole independently.

(a) Calculate the available upper- and lower-segment tolerances at the measured hole size. **[2 marks]**

(b) Assess both segments and hence the pattern. **[2 marks]**

(c) One hole is later found to measure only ⌀10.02. Reassess that hole if its required upper and lower zones are both the stated maxima, 0.46 and 0.22, while the common frameworks remain fixed. **[2 marks]**

(d) Describe the in-plane mobility permitted to the refined pattern by a lower segment that repeats only datum A. **[2 marks]**

(e) Explain why the lower segment controls hole-to-hole relationships more directly than the location of the pattern relative to B and C. **[2 marks]**

Level II

Question 14: Composite versus stacked position

Consider two alternative specifications for the same hole pattern.

Scheme 1: composite control

⌖	∅0.50 Ⓜ	A	B	C
	∅0.15 Ⓜ	A	B	

.

Scheme 2: two separate single-segment controls

⌖	∅0.50 Ⓜ	A	B	C
⌖	∅0.15 Ⓜ	A	B	

.

(a) Explain the principal difference between the lower composite segment and the second single-segment control. **[3 marks]**

(b) Which scheme better allows tight hole-to-hole spacing while retaining in-plane pattern float from both B and C within the upper-segment limits? Justify. **[2 marks]**

(c) Explain the effect of repeating datums A and B in the lower composite segment. **[2 marks]**

(d) State how the two schemes would differ during CMM evaluation. **[2 marks]**

(e) Give one manufacturing situation in which Scheme 2 would be intentionally preferred. **[1 mark]**

Question 15: Fixed-fastener assembly

Level II

A cover is attached to a base by fixed M8 fasteners. The fastener maximum shank diameter is ∅8.00, and the cover clearance holes have MMC size ∅8.50. The threaded-hole pattern in the base and the clearance-hole pattern in the cover are both position-controlled relative to functionally equivalent datum systems. Assume the fasteners follow the threaded-hole axes, their projected zones cover the cover thickness, and thread clearance, tilt, and assembly deformation are negligible.

(a) Derive the worst-case diametrical position-tolerance budget available to the two mating patterns. **[3 marks]**

(b) If the base pattern is assigned position ∅0.18, determine the maximum position tolerance that may be assigned to the cover at MMC. **[2 marks]**

(c) Assess a proposed allocation of ∅0.20 to the base and ∅0.35 to the cover. **[2 marks]**

(d) Explain how bonus tolerance may improve the acceptance of an individual assembly without increasing the worst-case MMC design budget. **[1 mark]**

(e) State the principal features, including pin diameter, of a functional gage for the 0.32 cover allocation from part (b). **[2 marks]**

Question 16: Floating-fastener assembly

Level II

Two plates are joined by a bolt that is free to float in clearance holes in both plates. Each hole has MMC size ∅8.50, and the bolt has maximum shank diameter ∅8.00. The patterns use functionally equivalent datum systems; assume a straight bolt, negligible tilt, and no other assembly constraints. Let the diametrical position tolerances assigned to the two hole patterns at MMC be T_1 and T_2.

(a) Derive the condition on $T_1 + T_2$ that guarantees worst-case assembly. **[3 marks]**

(b) Determine the maximum equal position tolerance that can be assigned to each plate. **[2 marks]**

(c) Assess the unequal allocation $T_1 = 0.60$ and $T_2 = 0.35$. **[2 marks]**

(d) For one assembly, the two actual hole sizes are ∅8.65 and ∅8.70. Determine the corresponding total position-error budget using the actual clearances. **[2 marks]**

(e) State the main reason this budget differs from that of a fixed-fastener assembly. **[1 mark]**

Question 17: Simultaneous requirement

Level II

Two separate hole patterns, P and Q, are controlled by position relative to the same datum reference frame $A|B(\text{MMB})|C$; datum feature B is referenced at MMB in both controls. Its available clearance permits every stated x shift. For this question, assume that a simultaneous requirement applies unless SEP REQT is stated. CMM analysis shows that pattern P can conform only when the common datum-simulator shift in the x direction lies in $[+0.08, +0.16]$, whereas pattern Q can conform only for a shift in $[-0.12, -0.04]$.

(a) Explain what the simultaneous requirement demands during functional inspection. **[2 marks]**

(b) Determine whether both patterns conform simultaneously. **[2 marks]**

(c) Explain why independent best fits for the two patterns are not valid under this requirement. **[2 marks]**

(d) Reassess the result if SEP REQT is added to both controls. **[2 marks]**

(e) Give one functional situation in which simultaneous control of the two patterns is necessary. **[2 marks]**

Question 18: Profile with datum targets

Level II

A thin cast housing is established on three primary datum targets A_1, A_2, A_3, two secondary targets B_1, B_2, and one tertiary target C_1. Its exterior is controlled by an equally disposed surface profile tolerance of 1.20 relative to $A|B|C$. CMM deviations, after alignment to the datum targets, are summarized below.

Region	Minimum deviation	Maximum deviation
R_1	-0.42	$+0.38$
R_2	-0.55	$+0.44$
R_3	-0.48	$+0.61$

(a) State the profile boundaries and assess each region. **[3 marks]**

(b) A global best fit produces deviations from -0.50 to $+0.49$. Explain why that result cannot automatically replace the datum-target evaluation. **[2 marks]**

(c) State how the 3–2–1 target arrangement constrains the six rigid-body degrees of freedom. **[3 marks]**

(d) Propose an additional control for a small sealing pad that requires profile accuracy substantially tighter than the rest of the casting. **[2 marks]**

Question 19: Runout diagnosis

Level II

A cylindrical surface is inspected relative to datum axis A with a common indicator zero. At three axial sections, the minimum, maximum, and mean radial readings over complete rotations are:

Section	Minimum	Maximum	Mean
Z_1	-0.010	$+0.030$	$+0.010$
Z_2	$+0.010$	$+0.050$	$+0.030$
Z_3	$+0.030$	$+0.070$	$+0.050$

The distance from Z_1 to Z_3 is 40 mm. For this simplified exercise, assume these sections include the global radial extrema of the entire controlled surface.

(a) Calculate circular runout at each section and assess a tolerance of 0.050. **[2 marks]**

(b) Calculate total runout over the complete surface and assess a tolerance of 0.060. **[2 marks]**

(c) Assuming approximately circular sections, use the change in mean radial reading to estimate the change in mean diameter from Z_1 to Z_3. **[2 marks]**

(d) Identify the likely axial form error suggested by the mean-value trend. **[2 marks]**

(e) Explain why circular runout alone can miss this error. **[2 marks]**

Question 20: Position versus runout for bores

Level II

A secondary bore must be coaxial with a primary datum bore B and correctly oriented to mounting face A. Two alternative controls are being considered:

⌖	∅0.040	A	B

or

⫽	0.040	B

.

CMM analysis reports a minimum required position-zone diameter of 0.036 for the derived axis. Indicator data on the bore surface give total runout 0.052 relative to datum axis B.

(a) Assess the bore under the position control. **[2 marks]**

(b) Assess the bore under the total-runout control. **[2 marks]**

(c) Explain which aspects of geometry are controlled by axis position but not directly by total runout, and vice versa. **[3 marks]**

(d) Select the more appropriate control for a close-running seal surface and justify the choice. **[2 marks]**

(e) State one additional form control that may be useful if the bore itself must be exceptionally round and straight. **[1 mark]**

Question 21: Sequential datum shift

Level III

A hole H has size ⌀8.00–⌀8.20 and position

⌖	⌀0.20 Ⓜ	A	B Ⓜ	C Ⓜ

.

Datum feature B is an internal slot of width 12.00–12.20 whose MMB is 12.00; its available shift acts only normal to the slot. Datum feature C is an internal cylindrical hole of size ⌀10.00–⌀10.20 with MMB ⌀10.00. The measured mating sizes are 12.14 for B and ⌀10.12 for C. For this simplified directional-shift model, use the full useful B shift in the y direction first, then the independent remaining useful C shift in the x direction; these quantities are stipulated rather than a general rule for all datum layouts. Hole H measures ⌀8.08 and has initial offsets $\Delta x = 0.18$, $\Delta y = 0.13$.

(a) Calculate the radial position-zone radius available for hole H. **[2 marks]**

(b) Calculate the available datum shifts from B and C in their useful directions. **[2 marks]**

(c) Calculate the residual x and y offsets after optimal use of the shifts and determine the residual diametrical position error. **[3 marks]**

(d) Decide whether the feature passes before and after datum shift. **[2 marks]**

(e) Explain why lower-precedence datum C cannot be used to undo a constraint already established by datum B. **[1 mark]**

Question 22: Composite position with varying sizes

Level III

Six holes have size ⌀6.00–⌀6.20. Datum A is the mounting plane; B and C are the in-plane locating datums. The pattern has the composite position control

⌖	⌀0.40 Ⓜ	A	B	C
	⌀0.15 Ⓜ	A		

.

From one admissible simultaneous fit of the common upper and lower frameworks to all six holes, the CMM reports the required zone diameters below. The holes have not been fitted separately.

Hole	Actual size (mm)	Upper zone required (mm)	Lower zone required (mm)
1	6.02	0.39	0.14
2	6.05	0.44	0.18
3	6.08	0.49	0.22
4	6.10	0.46	0.27
5	6.14	0.52	0.26
6	6.18	0.57	0.31

(a) For every hole, calculate the available upper- and lower-segment tolerances and record pass or fail for each segment. **[6 marks]**

(b) Identify the holes and segments that reject the pattern. **[2 marks]**

(c) State the acceptance decision for the complete pattern. **[1 mark]**

(d) Suggest one manufacturing correction targeted at the lower-segment failure. **[1 mark]**

Level III

Question 23: Functional gage design

A plate contains four holes of size ∅8.00–∅8.20 controlled by

⌖	∅0.20 Ⓜ	A	B Ⓜ	C

.

Datum A is a plane. Datum feature B is an external boss of size ∅20.00–∅20.10 with no separate geometric control; use MMB ∅20.10. Datum C is a side plane. One part has an actual mating size of ∅20.04 for datum boss B.

(a) Determine the diameter of each fixed gage pin used to check the four controlled holes. **[2 marks]**

(b) Determine the internal diameter of the datum feature simulator for boss B. **[2 marks]**

(c) Calculate the maximum radial mobility within the datum-B simulator for the stated actual mating size, before constraints from datum C and the controlled-hole gage pins. **[2 marks]**

(d) Describe the order in which datums A, B, and C are established in the functional gage. **[2 marks]**

(e) Explain why the gage pins are based on virtual condition rather than simply on the hole MMC size. **[2 marks]**

Question 24: Assembly over a temperature range

Level III

A fixed-fastener joint must assemble from 20°C to 100°C. At 20°C, the aluminium clearance hole has MMC diameter ⌀10.000 and the steel fastener has maximum shank diameter ⌀9.500. Use coefficients of thermal expansion $23 \times 10^{-6}/°C$ for aluminium and $12 \times 10^{-6}/°C$ for steel. Ignore thermal change of the basic pattern dimensions, fastener tilt, and thread clearance; assume equivalent assembly datums. The sum of the assigned diametrical position tolerances at MMC is 0.505.

(a) Calculate the worst-case assembly clearance budget at 20°C and assess the tolerance allocation. **[2 marks]**

(b) Calculate the minimum hole and maximum fastener diameters at 100°C. **[3 marks]**

(c) Determine the hot assembly-clearance budget and reassess the allocation. **[2 marks]**

(d) Explain why a design that passes only at the high-temperature condition is unacceptable for the stated requirement. **[1 mark]**

(e) If a design margin of 0.050 is required at 20°C and the base pattern uses position ⌀0.180, determine the maximum cover-pattern tolerance. **[2 marks]**

Question 25: Profile and machining stock

Level III

Five points on a forged surface are measured relative to the final true profile. Their outward deviations are

$$0.36, \quad 0.42, \quad 0.51, \quad 0.47, \quad 0.33.$$

A subsequent operation removes a uniform normal stock thickness s from the complete surface. The finished surface must satisfy an equally disposed profile tolerance of 0.30; hence every final deviation must lie between -0.15 and $+0.15$.

(a) Derive the interval of uniform stock-removal values s that makes all five points conform. **[4 marks]**

(b) Choose the value of s that centres the worst deviations within the tolerance zone. **[2 marks]**

(c) Calculate the five final deviations for the selected value and identify the maximum absolute deviation. **[2 marks]**

(d) Assess the process if $s = 0.50$ is used. **[2 marks]**

Question 26: Position of a slot

Level III

An internal slot has width 12.00–12.30. Its derived median plane is controlled by

⌖	0.20 Ⓜ	A	B	C

.

The slot measures 12.18. CMM evaluation shows that two parallel planes separated by 0.34, symmetrically disposed about the true centre plane, are required to contain the complete derived median plane.

(a) Calculate the bonus and total position tolerance for the slot. **[2 marks]**

(b) Assess the measured median-plane error. **[2 marks]**

(c) Calculate the virtual-condition width of the internal slot and the corresponding functional-gage tab width. **[2 marks]**

(d) Reassess the same median-plane error if the slot width were 12.05. **[2 marks]**

(e) Explain why no diameter symbol appears before the positional tolerance for this feature. **[2 marks]**

Question 27: Multiple datum reference frames

Level III

A housing uses base face A, locating side B, and end face C to establish the mounting datum reference frame. Bearing bore D is controlled by position $\varnothing 0.060$ relative to $A|B|C$. A seal counterbore is controlled by total runout 0.040 relative to datum axis D, and an adjacent seal face is controlled by an equally disposed surface profile tolerance of 0.080 relative to $D|A$. Inspection gives: bore-D position 0.052, counterbore total runout 0.036, and seal-face profile error 0.074, where the latter is twice the maximum absolute normal deviation from the true profile in its datum frame. For the additive estimate below, treat the counterbore as approximately cylindrical; use its runout value as a conservative diametrical surrogate for its axis offset relative to D.

(a) Identify the three datum reference sets used and the functional relationship established by each. **[2 marks]**

(b) Assess all three reported geometric errors. **[2 marks]**

(c) Using the stipulated additive estimate, calculate both the specified worst-case diametrical relationship budget and the estimate from the reported measurements for the counterbore axis relative to the basic bore-D location. **[2 marks]**

(d) Explain why controlling the counterbore directly to datum axis D may be functionally superior to locating it only from $A|B|C$. **[2 marks]**

(e) Outline a practical CMM inspection sequence that preserves the intended datum relationships. **[2 marks]**

Question 28: Conformance and capability

Level III

A hole has size ∅10.00–∅10.20 and position ∅0.20 at MMC. Five consecutive parts produce the following results. Define normalized tolerance utilization as

$$u = \frac{\text{required position-zone diameter}}{0.20 + \text{bonus tolerance}}.$$

Part	Actual hole size	Required position zone
1	10.01	0.18
2	10.04	0.25
3	10.08	0.24
4	10.12	0.29
5	10.18	0.35

(a) Calculate u for every part and state pass or fail. **[5 marks]**

(b) Identify the nonconforming part, if any. **[1 mark]**

(c) Identify the conforming part with the smallest remaining relative margin. **[2 marks]**

(d) Explain why a conventional capability index based on raw true-position values can mislead when bonus tolerance varies with size, and why five parts are insufficient for a long-term capability claim. **[2 marks]**

Question 29: Critique an over-controlled drawing

Level III

A preliminary drawing of a machined housing contains the following instructions:

1. Four mounting holes are located by both basic dimensions and ± 0.05 coordinate dimensions, and also have position $\varnothing 0.10$ relative to $A|B|C$.
2. Datum A is a small, rough as-cast pad, although the housing is mounted in service on a large machined base.
3. A bearing bore is controlled by the legacy concentricity symbol at 0.020 relative to an as-cast outside diameter designated datum D.
4. The machined base has flatness 0.050, parallelism 0.040 to datum A, and surface profile 0.030 to $A|B|C$ applied to the same complete surface.
5. The bearing-bore axis has both perpendicularity $\varnothing 0.030$ to A and position $\varnothing 0.020$ to $A|B|C$, with no stated functional reason for the looser orientation control.

(a) Identify and explain at least five specification, datum-selection, redundancy, or inspectability problems. **[5 marks]**

(b) Propose a corrected datum strategy based on the functional mounting interfaces. **[2 marks]**

(c) Replace the concentricity requirement with a more inspectable control suitable for a rotating or sealing relationship. **[1 mark]**

(d) Rewrite the mounting-hole location scheme without double tolerancing. **[1 mark]**

(e) State when a separate tighter orientation refinement on the bearing axis would be justified in addition to position. **[1 mark]**

Question 30: Bearing-housing capstone

Level III

A bearing housing has the following functional features:

- a large machined mounting base;
- one round locating hole and one relieved locating slot used with dowels;
- a bearing bore of size ∅50.000–∅50.025;
- a seal counterbore that must run concentrically with the bearing bore;
- a cover face normal to the bearing axis;
- six cover-bolt holes of size ∅8.50–∅8.80.

The functional requirements are: bearing-axis location and orientation within ∅0.040 of the mounting locating system; bearing-bore cylindricity 0.010; seal-counterbore total runout 0.030 relative to the bearing bore; cover-face total runout 0.030 measured axially over the full face relative to the bearing axis; bolt-pattern location ∅0.300 at MMC with hole-to-hole refinement ∅0.100 at MMC. Assume the bearing axis is basically normal to the mounting base, the slot's long direction lies along the line joining the two locating features, and the cover is located from the mounting datum system. Use RMB datum references for the strict mounting-axis location requirement.

(a) Select and order the primary, secondary, and tertiary datum features, using the mounting base, round locating hole, and relieved slot. Justify the order. **[2 marks]**

(b) Write suitable controls for the bearing-bore axis and bearing-bore form. **[2 marks]**

(c) Designate the bearing bore as a datum feature and write suitable controls for the seal counterbore and cover face. **[2 marks]**

(d) Write a composite positional control for the six bolt holes that meets the stated location and refinement requirements. **[2 marks]**

(e) Calculate the upper-segment virtual-condition diameter of the bolt holes. **[1 mark]**

(f) State one reason for using the relieved slot rather than a second round dowel hole as the tertiary locator. **[1 mark]**